Capacity is not redundancy: the assets that look strongest may not be the assets that keep the grid connected.

Question: If the utility can protect only a few assets, should it prioritize the most visible assets — high-capacity lines and high-degree buses — or the less obvious bottlenecks whose failure would isolate the most load or strand the most generation?

I use the phrase visible importance for things a typical observer could identify quickly from an asset table: line capacity, line length, bus degree, or total incident capacity. I use structural importance for a different property: whether an asset lies on many network paths and whether its removal fragments the grid, isolates load, or strands generation. The goal is not to simulate power flow; it is to learn what becomes visible when the grid is translated into a graph.

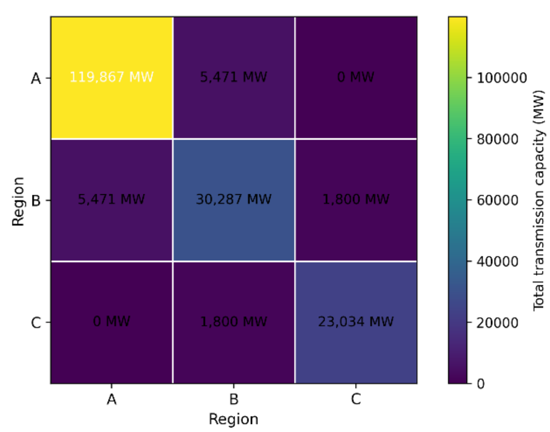
I framed the project around a practical maintenance question: if the utility could only protect a few assets, which ones should it choose? I wanted to test whether the obvious candidates, such as the highest-capacity lines or the most connected buses, were the parts of the grid whose failure would do the most damage. That led me to several smaller questions. If one transmission line failed, which outage would isolate the most load or strand the most generation? Does the answer come from the line's capacity, its length, or its position in the network? I also wanted to understand whether generation and demand are in the same places, or whether power must move across regions through a small number of corridors. From there, I looked at whether the grid has the same level of redundancy everywhere, or whether some areas are looped while others depend on single-feed structures. Finally, I compared degree, capacity, and betweenness to see whether they would recommend the same upgrade target.

I treated "big substations have more lines" as the obvious starting point, not the result I was trying to prove. In a real maintenance setting, those assets would already get attention because they are visible, expensive, and easy to justify protecting. What I wanted to test was whether that view misses quieter points of failure. My expectation was that some smaller or less-connected assets would turn out to matter more than they appear to, especially if they sit between where power is generated and where it is used. So, the main argument is that size and importance are not always the same thing. A high-capacity line may be important, but only the graph can show whether the system has another way around it. A modest bus may look ordinary, but if many routes depend on it, its failure can create much larger damage than its size suggests.

Network:

- **Nodes:** 300 buses on a regional electrical transmission grid, one row per bus, with node attributes including kind (generator, substation, or load), region (Control Area A, B, or C), capacity (MW), voltage (kV), and map coordinates.
- **Edges**: 422 weighted undirected edges. An edge between bus A and bus B represents a physical transmission line and carries a weight equal to its thermal carrying capacity (MW).
- **Data source:** Power Grid Project Dataset, a synthetic but realistic model of a fictional utility's transmission infrastructure.
- **Size:** N = 300, E = 422, density = 0.009. The largest connected component is the entire network — the power-grid is one connected regional electrical transmission system, with no isolated buses across its three control areas (generators, substations, and loads).

Figure 1. Region-pair transmission capacity



Note: The B-C cell is small relative to internal regional capacity, highlighting sparse but critical inter-region dependence.
Higher shading = larger total line capacity. Values are aggregated from the joined edge table.

Based on property of dataset and its application environments below are encodings I considered, evaluated and rejected. Here are the factual arguments for how to operate that network for a resilience analysis:

Directed vs. undirected. While electrical current flows in a specific direction at any given microsecond, the physical infrastructure of a transmission grid is designed to be undirected. Tie lines are built to carry power in either direction depending on where the generation is peaking and where the loads are drawing. I kept the network undirected because the resilience question is about whether a path exists at all between a generator and a load center. Treating the lines as directed would require a momentary snapshot of specific load balances that a static dataset cannot provide; an undirected approach honestly represents the structural redundancy of the physical wires.

Weighted vs. unweighted. An unweighted graph would treat a high-voltage, inter-regional trunk line the same as a small, local distribution lower-capacity line. Because the whole point of a grid analysis is to identify structural chokepoints, I needed the weights to distinguish between high-capacity corridors and minor ties. Without weights, degree centrality would collapse into a simple count of connections, incorrectly suggesting that a bus with three small lines is just as "important" as a bus with three massive trunk lines. Degree was computed on topology; strength and weighted betweenness used capacity.

Capacity (MW) vs. Line Length (km) as the weight. I considered weighting edges by their physical length (length_km) to represent impedance or voltage drop. However, I rejected this in favor of thermal carrying capacity (capacity_mw). In an engineering crisis, a line failure is a problem because it forces power onto neighboring lines; the critical limit is how many megawatts those neighbors can handle before they also trip. Capacity is the quantity an operator manages against to prevent structural exposure. A single 1,500 MW line provides more systemic stability than three 200 MW lines, and the weight choice should reflect that physical reality.

Because the grid is undirected, I created canonical endpoint pairs before aggregating. This means A-B and B-A are treated as the same region pair, and generator-substation is treated the same as substation-generator. Without this cleanup, the aggregation would split identical undirected relationships into separate rows just because of arbitrary endpoint ordering.

Procedure:

I started by loading nodes.csv, edges.csv, and regions.csv in Python. From those tables, I built an undirected weighted graph where each bus is a node, and each transmission line is an edge. Before running centrality, I checked the basic structure of the grid: total buses, total lines, connected components, density, and degree distribution. This confirmed that the grid began as one connected network but still had a sparse structure where individual bridges could matter.

The next step was joining node attributes onto the edge table. First, I attached the traits of the from_id endpoint to every transmission line. Then I repeated the join for the to_id endpoint, renaming those fields with clean to_ prefixes. After this double join, each line carried its own capacity and length, plus both endpoint regions, endpoint types, voltages, bus capacities, and labels. This was the key data-preparation step because it turned a raw line such as BUS0252 to BUS0161 into a meaningful system statement: a Region C substation connects to a Region B substation through 1,800 MW line.

Because the grid is undirected, I created canonical endpoint pairs before aggregating. A line listed as A-B and one listed as B-A represent the same regional relationship, so I sorted the endpoint labels into consistent region_pair and kind_pair fields. I then summarized the network by region pair and bus-kind pair, calculating line count, total capacity, mean capacity, maximum capacity, total length, and mean length. This moved the analysis from individual assets to the larger structure of the grid.

For centrality, I calculated degree, strength, unweighted betweenness, and capacity-weighted betweenness for every bus. Degree counts how many lines touch a bus, while strength sums the MW capacity connected to it. Unweighted betweenness shows how often a bus lies on fewest-hop paths. Capacity-weight betweenness uses 1 / capacity_mw as distance, so high-capacity corridors act as shorter paths. This does not mean electricity physically travels along graph shortest paths; it means the graph model treats high-capacity corridors as preferred structural routes for dependency analysis. I normalized betweenness to a 0-to-1 scale and rounded it to three decimals, so the results were easier to compare across buses and lines.

I also calculated line-level importance using capacity, length, unweighted edge betweenness, and capacity-weighted edge betweenness. This let me compare visible importance against structural importance: the biggest line, the longest line, and the line that acts most like a bridge in the network.

Finally, I tested criticality by removal. I removed every transmission line one at a time and recorded the largest connected component, number of components, isolated buses, isolated load MW, and stranded generation MW. I also removed the top assets under competing rules: highest degree, highest incident capacity, highest bus betweenness, highest line capacity, highest line length, and highest edge betweenness. I compared those targeted removals with random-removal baselines using a fixed seed. That comparison matters because it shows whether the selected assets are structurally critical, rather than just harmful because any removal would damage the grid.

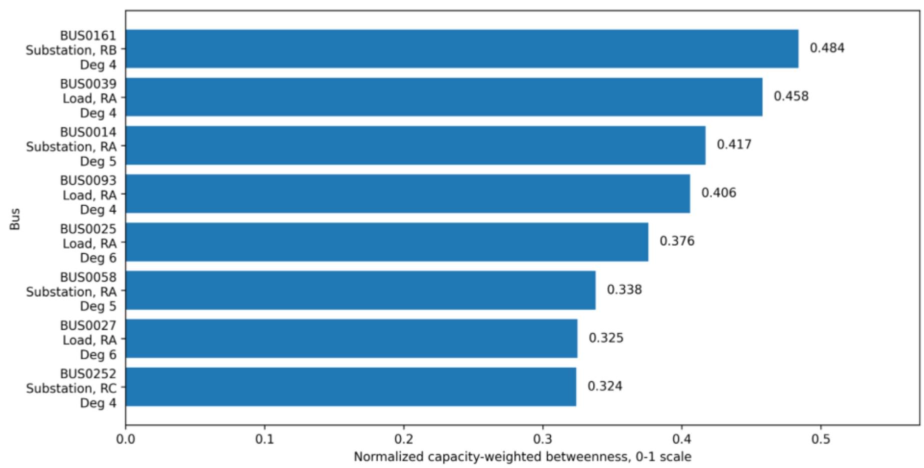
Results:

I organize the results around the protection decision: if the utility can protect only a few assets, the question is whether those assets should be the most visible ones or the less obvious bottlenecks. The first finding is that power is made and used in different places.

Regions A and B are load-heavy, with 20,999 MW of load and only 5,352 MW of generation, while Region C is generation-heavy, with 20,667 MW of generation and only 254 MW of load. That mismatch makes the transmission structure critical, because power must move from Region C toward the demand side of the system. This regional imbalance is summarized in the generation-load table and visually supported by **Figure 1**, this is why the inter-region ties, not the regional totals, become the binding constraint.

The regional structure shows that the grid is not equally redundant. Region A has strong internal mesh, Region B has radial tails, and Region C is externally exposed. The key weakness is the inter-region structure: B-C has only one 1,800 MW tie, and there is no direct A-C tie. This means the B-C line is not just a large asset; it is a bridge between the supply-heavy region and the load-serving side of the network. A highly capable corridor can still be fragile if there are few alternatives around it.

Figure 2. Hidden Bottleneck Buses by Normalized Capacity-Weighted Betweenness

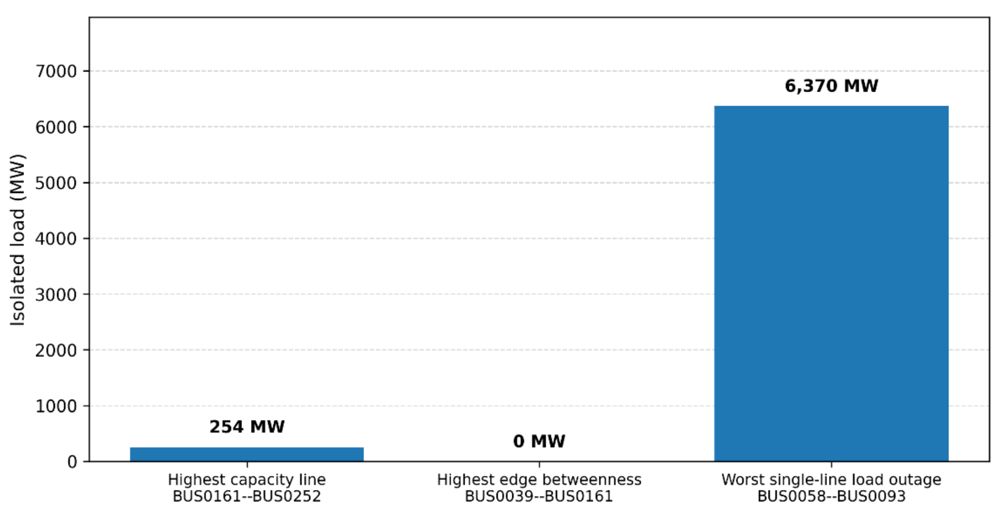


Betweenness is normalized and rounded to three decimals. Higher values indicate buses used by more capacity-preferred shortest paths.

The bus-level results show the same pattern. BUS0161 is the clearest hidden bottleneck. It is only a degree-4 substation with 0 MW bus capacity, so it does not look like the biggest or most connected asset. But it has the highest capacity-weighted betweenness in the network, which means many important paths depend on it. When the top five buses by degree are removed, the grid remains connected. When the top five buses by betweenness are removed, the grid splits into 5 components, with 6,624 MW of load and 21,784 MW of generation outside the largest component.

The line-outage results make the protection problem even sharper. If the goal is to avoid isolated load, the worst single-line outage is BUS0058-BUS0093, which isolates 6,370 MW of load even though it is only a 356 MW line. If the goal is to avoid stranded generation, the worst line is BUS0161-BUS0252, the 1,800 MW B-C intertie, which strands 20,667 MW of generation. So, there is no single answer to “which line matters most” unless the damage metric is defined.

Figure 3. Single-line outage damage by selection rule

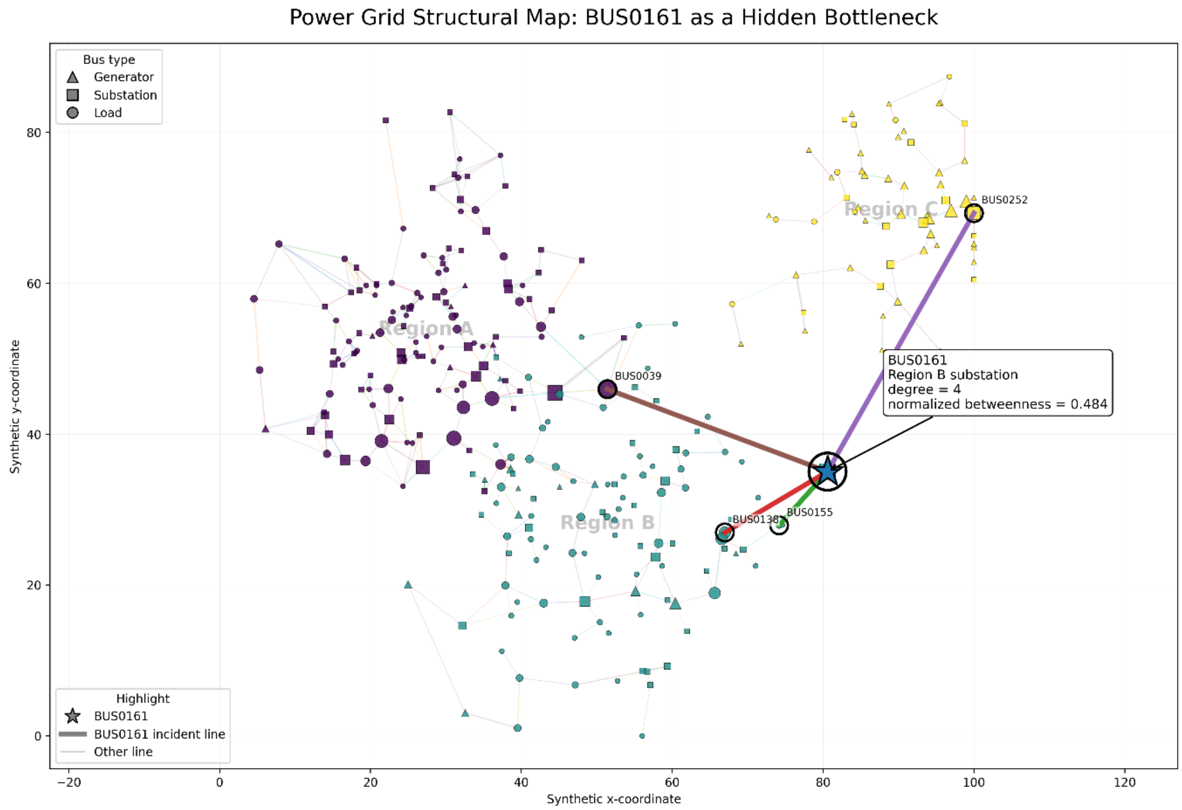


Capacity alone points to the B-C export intertie; outage impact points to the smaller BUS0058--BUS0093 load-continuity bottleneck.

Overall, the results show that degree, capacity, betweenness, and outage impact do not recommend the same protection target. Degree flags local hubs and capacity flags the big transfer assets — but neither predicts what betweenness shows about structural dependency, and outage testing is really the only one that tells you the actual consequence of losing something. The practical lesson is that the utility should not rely on visible importance alone. Capacity is important, but capacity is not redundancy.

What the analysis reveals --- and what it does not;

The analysis shows that a utility should not protect assets based on visible importance alone. The highest-capacity lines and highest-degree buses matter, but they do not fully identify the failures that would isolate the most load or strand the most generation. Capacity-weighted paths reveal the main transfer structure, while betweenness and outage testing show where that structure is most dependent on a few buses or lines. This is especially important because Region C produces most of the power, while Regions A and B use most of it, making the B-C corridor and other inter-region links structurally important. The analysis is not an AC power-flow model and does not include voltage, dispatch, relay action, overload timing, weather, or hourly demand, so it identifies structural exposure rather than a full dynamic cascade. Still, that is enough for the main lesson: BUS0161 makes that concrete: it's a nothing-looking degree-4 substation, yet the paths running through it are the ones that hold the grid together. The utility should protect alternatives and bottlenecks, not just the biggest assets.



All 300 buses and 422 transmission lines shown. Node size scales with normalized capacity-weighted betweenness; edge width scales with line capacity. BUS0161 and its direct incident lines are highlighted.